PROJECT PROGRESS REPORT 1

Automated Classification of Sports-Related Injuries via Medical Imaging

Reporting Period: Week 1–2 | Date: April 5, 2026 | Phase: Setup & Data Acquisition

Task	Status	Notes
Project scoping & proposal finalization	COMPLETE	All objectives confirmed
Dataset identification (MURA / Kaggle)	COMPLETE	Primary & fallback sources secured
Environment setup (Python, PyTorch, libraries)	COMPLETE	GPU environment configured
Initial data download & file integrity check	COMPLETE	~40,000 X-ray images acquired
Preprocessing pipeline design	IN PROGRESS	Resize + normalization implemented
Exploratory Data Analysis (EDA)	PLANNED	Begins Week 3

1. Overview

This report documents progress for Weeks 1 and 2 of the sports injury classification project. The primary focus during this phase was establishing the project infrastructure, sourcing an appropriate medical imaging dataset, and designing the data preprocessing pipeline. All critical setup milestones have been completed on schedule.

2. Data Acquisition

2.1 Dataset Selected

After evaluating multiple options, the **MURA (Musculoskeletal Radiographs) dataset** from the Stanford ML Group was selected as the primary data source, supplemented by the Kaggle Bone Fracture Detection Dataset as a fallback and augmentation resource.

Dataset	Images	Classes	Anatomy Sites
MURA (Stanford)	~40,000	Normal / Abnormal	Elbow, Shoulder, Wrist, Knee
Kaggle Fracture Set	~10,000	Fracture / Non-Fracture	Hand, Wrist, Femur

2.2 Data Integrity Verification

Following download, all files were verified for integrity. Key findings include:

- Total images downloaded: **40,561** (MURA) + **9,876** (Kaggle)

• Corrupt/unreadable files identified and removed: **43 files**

• Confirmed all images are anonymized per dataset provider's IRB agreement

• Image formats: primarily JPEG and PNG at varying resolutions (ranging 200×200 to 1200×1000 px)

3. Preprocessing Pipeline (In Progress)

The preprocessing pipeline is currently being implemented in PyTorch using the **torchvision.transforms** module. Steps completed and in-progress are detailed below:

• **Image Resizing** [DONE]: All images standardized to 224×224 pixels for ResNet50/MobileNetV2 compatibility.

• **Normalization** [DONE]: Pixel values scaled to [0, 1] range; ImageNet mean/std applied for transfer learning alignment.

• **Augmentation** [IN PROGRESS]: Random horizontal flips, ±15° rotations, and brightness jitter (factor 0.2) being tested.

• **One-Hot Encoding** [PLANNED]: Label encoding to be finalized once EDA class distribution analysis is complete.

4. Development Environment

Component	Details
Framework	PyTorch 2.2 + torchvision
Python Version	3.11
Hardware	NVIDIA RTX 3090 (24GB VRAM)
Key Libraries	NumPy, Pandas, Matplotlib, Scikit-learn, Pillow
Version Control	Git (private repository)

5. Next Steps (Week 3–4)

• Complete augmentation pipeline and run validation on sample batch of 500 images.

• Perform full Exploratory Data Analysis: class distribution charts, intensity histograms, and sample grids.

• Finalize One-Hot Encoding and split dataset into 70/15/15 train/validation/test sets.

• Begin baseline CNN architecture design (3–4 convolutional layers).

• Document EDA findings in Progress Report 2.